Completely Contactless and Online Finger Knuckle Identification for Real World Applications

Zhenyu Zhou, Ajay Kumar

Abstract

Completely contactless and at-a-distance personal identification provides enhanced user convenience, and improved hygiene and is highly sought under the COVID-19 pandemic. This paper proposes an accurate and generalizable deep neural network-based framework for the 'completely' contactless finger knuckle identification. We design and introduce a new loss function to enable a fully convolutional network to more effectively learn knuckle features that are imaged under at-a-distance imaging. A 'completely' contactless system also requires efficient online finger knuckle detection capabilities. This paper, for the first time in our knowledge, develops and introduces accurate capabilities to efficiently detect and segment finger knuckle patterns from images with *complex* backgrounds as widely observed in real-world applications. We introduce angular loss to accurately predict oriented knuckle patterns and incorporate into our framework. Experimental results presented in this paper on five different public databases, using challenging protocols and cross-database performance evaluation, illustrate outperforming results and validate the effectiveness of the proposed framework for completely contactless applications.

Index Terms

Biometrics, Completely Contactless Finger Knuckle Identification, At-a-distance Finger Knuckle Detection

I. Introduction

AMONG a range of physiological characteristics for biometrics, finger knuckle patterns have the least privacy concerns, are convenient to be imaged from a distance, are easier to be presented, and more importantly, they have shown to offer high uniqueness and stability among the adult population [9]. Contactless or at-a-distance biometrics systems have high user acceptance [1], are highly preferred under the ongoing pandemic, and are effective in preventing cross-contamination [7] during COVID-19 pandemic. Finger knuckle is a relatively new biometric and has attracted increasing attention in the forensics and biometrics community.

There have been earlier studies and work on contactless finger knuckle biometrics. However, in the context of user convenience and hygiene, it is important to differentiate the contactless and 'completely' contactless finger knuckle systems. Contactless biometrics systems ensure that there is no physical contact between the sensor and the imaged biometric. However, this has often included the systems [2], [3], [6], [8], [10], [11] that require the user to present his/her fingers or hands on a surface. On the other hand, a 'completely' contactless finger knuckle system will ensure that neither the fingers nor hands are presented in the completely contactless manner [5] in the field of view of the sensor which itself can be mobile. Such a completely contactless system can allow 6 degrees of freedom during the imaging, and acquire images with complex background patterns with a high degree of rotational and scale changes. Therefore, the framework for the *completely* contactless system should ensure accurate detection (where the position, size, and the number of finger knuckles appearing in each frame can vary) and segmentation of knuckle patterns under complex backgrounds, and provide higher accuracy under such relatively higher intra-class variations. This work attempts to address such open challenges and realize a completely contactless and online system for real-world deployments.

Authors are with the Department of Computing, The Hong Kong Polytechnic University, Hung Hom, Kowloon, Hong Kong.

A. Related Work

Contactless Finger Knuckle Identification: There have been a several exciting attempts to match contactless finger knuckle images, e.g. using a range of methods that that attempted to incorporate key points [12], local spatial features using binary representations [35], subspace methods [13] to encode local 3D curvature of knuckle creases, and illustrate impressive performance. Reference [4] provides a comprehensive summary of such methods. Even earlier work has considered application scenarios using smartphones for the contactless finger knuckle recognition [14], but the finger knuckle segmentation requirement to present the finger in a fixed imaging position i.e. image center, is inconvenient and limited its practical use.

Conventional methods heavily rely on the selection of parameters when incorporated for matching the images for the other/different databases or those acquired under different imaging setups. On the contrary, deep neural network-based methods can offer enhanced generalization and robustness and have been recently attempted in the literature. Reference [8] presents FKNet which is based on the ResNet-50 [15], and can provide robust deep feature representation for the finger knuckle features. Attempts to combine traditional methods with CNN models appear in [16] which firstly use initialized Gabor filters to extract features, and then use CNN to learn the most important information, and the Gabor filter parameters can also update during backpropagation. The models in [8], [16] are complex and lightweight models, like the recently introduced EfficientNetV2 [17] model, can provide a better alternative for real-time deployments.

Contactless Fingers Detection and Segmentation: The accuracy of the detected and segmented region of interest (ROI) can determine the matching efficiency and accuracy of the online finger knuckle system. The size of the ROI should be comparable to the size of the actual target object (knuckle) at the pixel level so that the size obtained after segmentation will not have redundant pixel information, which will reduce the pixel values to be computed for both the extraction and the subsequent matching sessions.

Conventional contactless finger knuckle methods, their [18], [14] approach is to fix the finger knuckle position in the image while acquiring the finger knuckle images under plain, instead of the complex background, and then detect the edges of the finger, and then extract the finger knuckle crease regions. Prior finger knuckle detection and segmentation algorithm cannot correctly detect/segment the finger knuckles in the presence of complex backgrounds, multiple finger knuckles in the *same* field of view, obscured or finger knuckles regions with high pose and scale changes.

Deep neural network models for object detection have achieved great success, whether it is the sliding window detection algorithm, the 2-stage series of R-CNN models [19], [20], [21], or the 1-stage YOLO series [22], [23], [24], [25] and SSD models [26] up to the current position, and even the anchor-free based object detection algorithm [27] as well. Each of these models has its advantages and limitations. For the 2-stage model, the object detection accuracy is quite guaranteed, the 1-stage based model is a speedup based on the positive accuracy, and the anchor-free is a further improvement in the detection speed. In this paper, we adopt a recent version of the YOLO model, i.e. YOLOv5 [28], for the fast and accurate detection of finger knuckle regions.

Limitations and Challenges: As outlined in previous sections, conventional methods for finger knuckle identification require parameter tuning which is specific to the database and even the parameters for the detection should be tuned for different imaging environments. Deep neural network architectures are known to offer high generalization capabilities. However, any direct application of such architectures, e.g. [8], [16], [29] is expected to deliver *limited* performance. This is due to the fact that new challenges emerge while incorporating completely contactless finger knuckle images. Earlier work [8], [16], [29] have fail to effectively accommodate inherent rotational and translational changes that are quite common under completely contactless imaging. Also, the number of output classes is fixed in [8] for the training, and therefore it requires retraining whenever a new subject is added.

The finger knuckle detection and segmentation approach used in [35] cannot work for completely contactless imaging under complex background. The knuckle detection and segmentation approach in [8] is based on the Mask R-CNN which is highly complex and not suitable for the real-time identification

TABLE I
COMPARATIVE SUMMARY OF RELATED AND RECENT WORK ON CONTACTLESS FINGER KNUCKLE IDENTIFICATION

Ref.	Deployment Scenario			Performance		
	Real-World Knuckle Detection	Completely Contactless	Online System	Database	EER	GAR @ FAR=10^{-4}
[8]	No	No	No	A	3.90%	86.60%
				B	7.70%	74.50%
				D(1)	2.40%	94.70%
[11]	No	No	No	A	2.72%	N/A
				E	0.66%	N/A
[29]	No	No	No	A	3.97%	30.00%
[10]	No	No	No	D(1)	9.60%	81.50%
				D(2)	10.20%	76.00%
Ours	**Yes**	**Yes**	**Yes**	B	**1.33%**	**96.07%**
				C	**2.21%**	**91.48%**
				D(1)	**1.05%**	**97.10%**
				D(2)	**1.60%**	**96.05%**
				E	**0.16%**	**99.80%**

Database description: A: PolyU Contactless Finger Knuckle Images Database (Version 1, 501 subjects) [30]; B: PolyU Contactless Hand Dorsal Images Database (712 subjects) [31]; C: PolyU Contactless Finger Knuckle Images Database (Version 3, 221 subjects) [32]; D: PolyU 3D Finger Knuckle Images Database (D(1) 3D Images, D(2) 2D Images, 190 subjects) [33]; E: Tsinghua University Finger Vein and Finger Dorsal Texture Database (610 subjects) [34].

system. At present, popular deep neural network-based object detection and segmentation methods cannot simultaneously recover the orientation angle of finger knuckle and segment knuckle region with high precision. However, accurate estimation of the presented finger knuckle is a key factor for the success of online finger knuckle identification.

B. Our Works and Contributions

Completely contactless and online finger knuckle identification is a difficult problem, largely due to the complex capabilities required to match deformable knuckle crease patterns that are often acquired/presented from the varying viewing angle. In order to accurately and robustly match such patterns, a deep learning-based framework that can recover rotational and translational invariant features should be designed. Another open problem in building a completely contactless finger knuckle identification system is related to accurate detection and segmentation of the ROI under rotational, translation, and scale changes. Conventional methods in the literature for knuckle detection and segmentation are inadequate and cannot meet such expectations for completely contactless systems. Table I summarized key advantages of our system and the related system.

This paper introduces an accurate and robust contactless finger knuckle feature representation framework and develops *completely* contactless finger knuckle identification capabilities for at-a-distance real-world applications. It makes significant contributions toward discovering full potential from the deep learning capabilities and such objectives are yet to be pursued in the biometrics literature.

Key contributions from this paper can be summarized as follows:

- We develop a new deep learning-based contactless finger knuckle identification framework with high generalization capability for operating on different contactless finger knuckle databases. A new *Rotation and Shift Invariant Loss* (RSIL) function has been developed to address the degradation in matching performance due to high intra-class variations among knuckle crease patterns. The proposed loss is generalized and also has the potential to be employed in many other contactless matching tasks. Outperforming experimental results presented in section IV-A to IV-E of this paper using five different contactless finger knuckle databases, under challenging protocols and using cross-database performance evaluation, validate the effectiveness of the proposed framework for contactless finger knuckle identification.
- We also design and develop a completely contactless finger knuckle detector to accurately detect, segment, and predict respective knuckle orientations from the multiple hands in the real-world images.

We use Yolov5 [28] as the backbone with circular smoothing labels (CSL) [36], and introduce *angular loss* to predict the oriented bounding box to accurately segment the ROI with knuckle patterns. Our experimental results in section IV-F to IV-G of this paper also validate the effectiveness of angular loss to enhance the performance for the completely contactless finger knuckle detection.

- We design a cross-platform online finger knuckle identification system for completely contactless and real-world applications. The system can detect and match finger knuckle patterns from finger images acquired under complex background, using any ordinary smartphone or general laptop camera. It's the first completely contactless and online system (please see *attached* video samples recorded from its live operation), to the best of our knowledge, and a significant step to avail full potential of knuckle biometrics for social good.

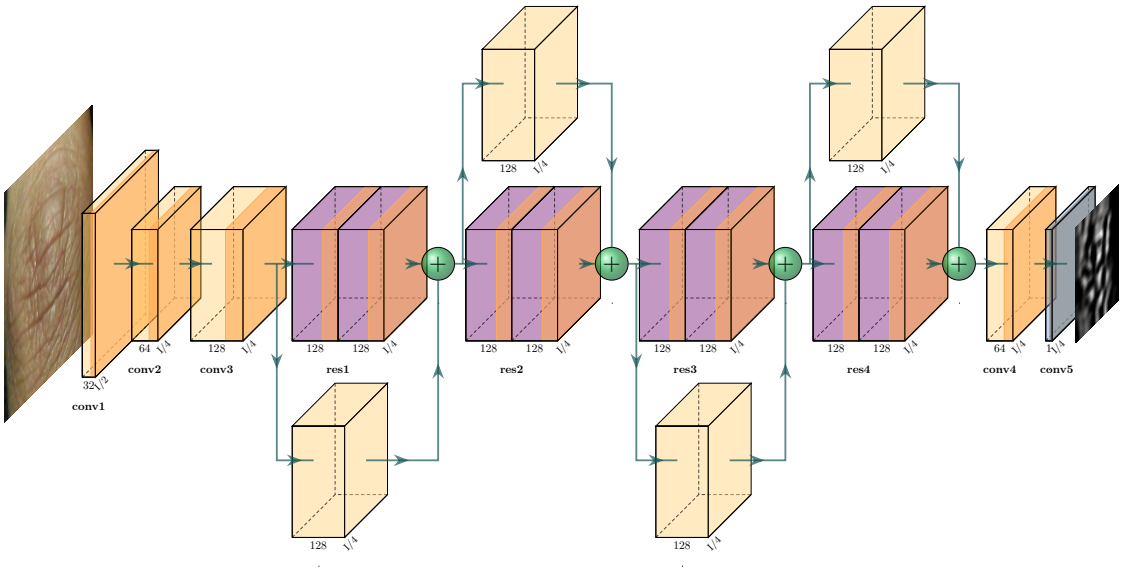


Fig. 1. Architecture of a fully convolutional network employed as the backbone in our framework.

The rest of this paper is organized as follows: In section II we introduce our deep learning framework for completely contactless finger knuckle matching. This section includes the details on the proposed loss function and its differentiability requirements for the network training. Section III introduces developed solution to accurately detect the finger knuckle ROI and their respective orientations, from the completely contactless hand dorsal images with complex background. Section IV presents all the experimental results relating to the identification including within-database experiments and cross-database experimental results, and the comparative performance for the 3D finger knuckle database. This section also includes performance evaluation from a completely contactless finger knuckle identification developed during this work. Finally, the key conclusions from this work are summarized in section V of this paper.

II. Matching Completely Contactless Finger Knuckle Images

Most relevant information from the knuckle crease patterns can be described using the inter-pixel relationship among the gray-level arrangements in the segmented images. Such a relationship is expected to be better *preserved* in the output templates generated by a fully-convolutional network output, instead of the fully connected output as in *prior work* [8]. This is the motivation for choosing a lightweight residual network [37] with fully convolutional output, which is also referred to as a residual feature network (RFN), as the backbone for matching contactless knuckle images. Fig. 1 illustrates backbone architecture for the contactless finger knuckle matching. We use triplet architecture [38] to train the convolutional kernels of this network. Three such shared networks are interconnected in parallel to enable

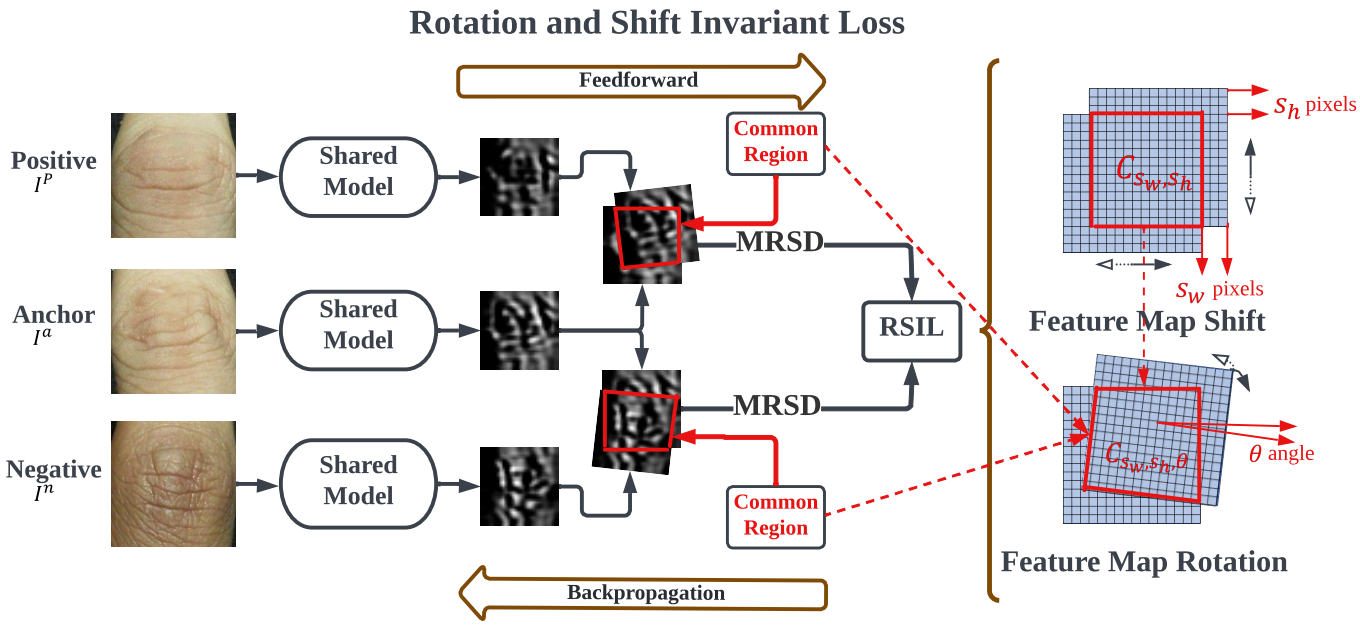


Fig. 2. An overview to illustrate the use of rotation and shift invariant loss function (RSIL) to train the network under triplet architecture. We will use MRSD to compute the similarity between two output templates (feature maps shown on the right). The common knuckle region is illustrated with red boundary after accommodating for potential rotational and translational changes.

forward and backward propagation of the gradients of network loss during the offline training. Completely contactless and normalized finger knuckle images often illustrate significant rotational misalignments and deformations among the curved knuckle creases. Therefore, conventional triplet loss or shifted triplet loss is not adequate to address such rotational and translational changes. We therefore design a new loss function which can enable the network to effectively learn the most robust features that are also rotational invariant due to multiple degrees of freedom during the imaging. This new loss function is detailed in the following section.

A. Rotation and Shift Invariant Loss Function

In this section, we systematically introduce a new loss function RSIL and desire such requirement for its differentiability. With a differentiable loss, the backpropagation process can proceed smoothly, and the learnable parameters can be updated to until RSIL is reduced to some minimum value. Our backbone network is trained using the triplet architecture for which the conventional triplet loss [38] can be stated as in the following:

$$TL = \frac{1}{N}\sum_{i}^{N}[||T(I_i^a) - T(I_i^p)||^2 - ||T(I_i^a) - T(I_i^n)||^2 + m]_+ \tag{1}$$

where N designates the batch size employed during training iteration; $T(I^a), T(I^p), T(I^n)$ respectively represent the output feature matrix (template) generated by the network from the input anchor, positive and negative ROI image I^a, I^p, and I^n, The symbol $[\circ]_+$ represents the same function as $max(\circ, 0)$ and is used to ensure a non-negative loss. The hard margin parameter m can effectively determine the distance between different class clusters by pushing them away during the training process.

The triplet loss computed in (1) is however not appropriate for matching *completely* contactless finger knuckle images. Normalized contactless finger knuckle images from the same client's fingers will not only illustrate the spatial translation of knuckle crease patterns but are often accompanied by rotations which

can be due to the involuntary local deformations while presenting the fingers for the contactless imaging. In order to robustly accommodate such intra-class changes in the contactless finger knuckle patterns, we alter the original triplet loss function and introduce RSIL as follows:

$$RSIL = \frac{1}{N}\sum_{i}^{N}[L(T(I_i^a), T(I_i^p)) - L(T(I_i^a), T(I_i^n)) + m]_+ \tag{2}$$

In order to adapt to the intra-class knuckle patterns with varying degrees of rotational and translational changes, and balance the performance and speed, the translational and rotational parameters are set as the hyperparameters. The $L(T_1, T_2)$ in (2) computes the minimum distance between the two templates $D_{s_w,s_h,\theta}(T_1, T_2)$ after accounting for shift and rotation in the range of $-S_W \leq s_w \leq S_W$, $-S_H \leq s_h \leq S_H$, $-\Theta \leq \theta \leq \Theta$, called minimal rotational and shifted distance (MRSD) in (3). We use S_W, S_H represent the extent of horizontal and vertical pixel translations respectively, and Θ for rotaion angle. The distance $D_{s_w,s_h,\theta}(T_1, T_2)$ computes the pixel-wise MSE value when template T_1 is shifted s_w pixel along x-axis and s_h pixel along y-axis and rotated by θ angle as also shown in Fig. 2.

$$L(T_1, T_2) = \underset{\substack{-S_W \leq s_w \leq S_W, \\ -S_H \leq s_h \leq S_H, \\ -\Theta \leq \theta \leq \Theta}}{min} D_{s_w,s_h,\theta}(T_1, T_2) \tag{3}$$

$$D_{s_w,s_h,\theta}(T_1, T_2) = \frac{1}{|C_{s_w,s_h,\theta}|}\sum_{(x,y)\in C_{s_w,s_h,\theta}} (T_1^{(s_w,s_h,\theta)}\lfloor n, m \rfloor - T_2[x, y])^2 \tag{4}$$

where $C_{s_w,s_h,\theta}$ represents the common region between two feature maps after one feature map is translated along the x-axis by s_w, translated along y-axis by s_h pixels, and rotated by θ degrees, as shown in the Fig. 2. In (4), the $\lfloor \circ \rfloor$ is the operator for conventional floor function. When feature maps are rotated and shifted, using affine transformation, the new feature map $T^{(s_w,s_h,\theta)}\lfloor n, m \rfloor$ entries can be directly recovered from the source $T[x, y]$ depending on the sampling method as follows:

$$T^{(s_w,s_h,\theta)}\lfloor n, m \rfloor = \sum_{y}^{H}\sum_{x}^{W} T[x, y]p(n - x)q(m - y) \tag{5}$$

where the height and width of $T[x, y]$ is represented by H and W, while $[x, y]$ represents the original pixel positions *before* the affine transformation; $p()$ is the interpolation operation along the x-axis while $q()$ is the interpolation operation along the y-axis. Since use bilinear interpolation, the above equation can be rewritten as follows:

$$T^{(s_w,s_h,\theta)}\lfloor n, m \rfloor = \sum_{y}^{H}\sum_{x}^{W} T[x, y]max(0, 1 - |n - x|)max(0, 1 - |m - y|) \tag{6}$$

The training of network will require us to compute the gradients, or the partial derivatives, of RSIL which will be iteratively backpropagated during the training. We can firstly compute the partial derivative of $T^{(s_w,s_h,\theta)}\lfloor n, m \rfloor$ with respect to $T[x, y]$ as in the following.

$$\frac{\partial T^{(s_w,s_h,\theta)}\lfloor n, m \rfloor}{\partial T[x, y]} = \sum_{y}^{H}\sum_{x}^{W} max(0, 1 - |n - x|)max(0, 1 - |m - y|) \tag{7}$$

As for the (T^a, T^p) pair, we can assume when the T^a is rotated by angle of θ_{ap} and shifted with (s_{wap}, s_{hap}) pixels, the minimum distance $D_{s_{wap},s_{hap},\theta_{ap}}(T^a, T^p)$ can be achieved. Therefore the resulting loss $L(T^a, T^p) = D_{s_{wap},s_{hap},\theta_{ap}}(T^a, T^p)$. Similarly, for the (T^a, T^n) pair, with the alignment by $s_{wan}, s_{han}, \theta_{an}$,

minimum distance $D_{s_{wan},s_{han},\theta_{an}}(T^a, T^n)$ can be computed. We can then compute the partial derivation of RSIL with respect to T_i^p and T_i^n, respectively as follows:

$$\frac{\partial RSIL}{\partial T_i^p} = \begin{cases} 0, & if\ (x,y) \notin C_{s_{wap},s_{hap},\theta_{ap}}\ or\ RSIL = 0 \\ -\dfrac{2(T_i^{a(s_{wap},s_{hap},\theta_{ap})}\lfloor n,m \rfloor - T_i^p[x,y])}{N \times |C_{s_{wap},s_{hap},\theta_{ap}}|}, & otherwise \end{cases} \tag{8}$$

$$\frac{\partial RSIL}{\partial T_i^n} = \begin{cases} 0, & if\ (x,y) \notin C_{s_{wan},s_{han},\theta_{an}}\ or\ RSIL = 0 \\ \dfrac{2(T_i^{a(s_{wan},s_{han},\theta_{an})}\lfloor n,m \rfloor - T_i^n[x,y])}{N \times |C_{s_{wan},s_{han},\theta_{an}}|}, & otherwise \end{cases} \tag{9}$$

where $C_{s_{wap},s_{hap},\theta_{ap}}$ represents the common region between the anchor and positive sample, while $C_{s_{wan},s_{han},\theta_{an}}$ represents respective common region between anchor and negative sample after such rotation and translation. Translation and rotation anchor template will generate respective *interpolated* templates, $T_i^{a(s_{wap},s_{hap},\theta_{ap})}$ and $T_i^{a(s_{wan},s_{han},\theta_{an})}$. The partial derivative of T_i^a can be computed as follows:

$$\frac{\partial RSIL}{\partial T_i^a} = \begin{cases} 0, & if\ (x,y) \notin C_{apn}\ or\ RSIL = 0 \\ \dfrac{2(T_i^{a(s_{wap},s_{hap},\theta_{ap})}\lfloor n,m \rfloor - T_i^p[x,y])}{N \times |C_{s_{wap},s_{hap},\theta_{ap}}|} \times K + \\ \dfrac{-2(T_i^{a(s_{wan},s_{han},\theta_{an})}\lfloor n,m \rfloor - T_i^n[x,y])}{N \times |C_{s_{wan},s_{han},\theta_{an}}|} \times L, & otherwise \end{cases} \tag{10}$$

where *K* and *L* respectively represents the partial derivation of $\partial T_i^{a(s_{wap},s_{hap},\theta_{ap})}\lfloor n,m \rfloor / \partial T_i^a[x,y]$ and $\partial T_i^{a(s_{wan},s_{han},\theta_{an})}\lfloor n,m \rfloor / \partial T_i^a[x,y]$ which can be computed using (7), and C_{apn} represents common region between tow spatial regions i.e. $C_{s_{wan},s_{han},\theta_{an}} \cup C_{s_{wap},s_{hap},\theta_{ap}}$ as C_{apn}. The RSIL loss computed using the intermediate template. T_i^a, T_i^p and T_i^n will be backpropagated using the chain rule to update the learnable parameters of neural network during the training.

III. Finger Knuckle Detection in a Wild

Real-world deployment of *completely* contactless finger knuckle identification capabilities will require accurate detection of the such ROI from a complex background that is often expected. There are several possibilities to train a deep network based popular object detectors, e.g. YOLO [22], [23], [24], [25], [28], Mask R-CNN [39], to detect and segment knuckle ROI. Such attempts have also been recently used in the literature, e.g. [8] uses Mask R-CNN [39] to detect and segment Knuckle ROIs. In this context, we can note that the YOLOv4 [25] object detection model has shown to be superior i.e., the fastest and most accurate detectors in terms of both speed and accuracy when compared to other state-of-the art object detection model. However, such methods cannot predict the orientation of knuckle ROI which is indeed required to accurately match *completely* contactless knuckle images. Therefore, accurate prediction of the orientation of knuckle ROI is desirable to align two feature maps for enhancing the accuracy and efficiency. The YOLO series model can be easily adapted for predicting *oriented* bounding boxes. We, therefore, incorporate circular smooth labels to smooth the prediction of knuckle orientation classes using YOLOv5 [28] model. Our approach for predicting oriented bounding boxes corresponding to ROI under complex background images is detailed in the following.

The complexity and detection speed offered by YOLOv5 can meet expectations for inference time and accuracy. However, a trained YOLOv5 model can just detect horizontal bounding boxes which cannot offer orientation information and will also segment *unwanted* background details. Therefore, to address such challenges for our problem, an oriented bounding box is predicted instead of the conventional or horizontal bounding box. The loss for the oriented bounding boxes is largely contributed from the angular periodicity loss, along with exchangeability of edges loss, when bounding boxes are defined using five

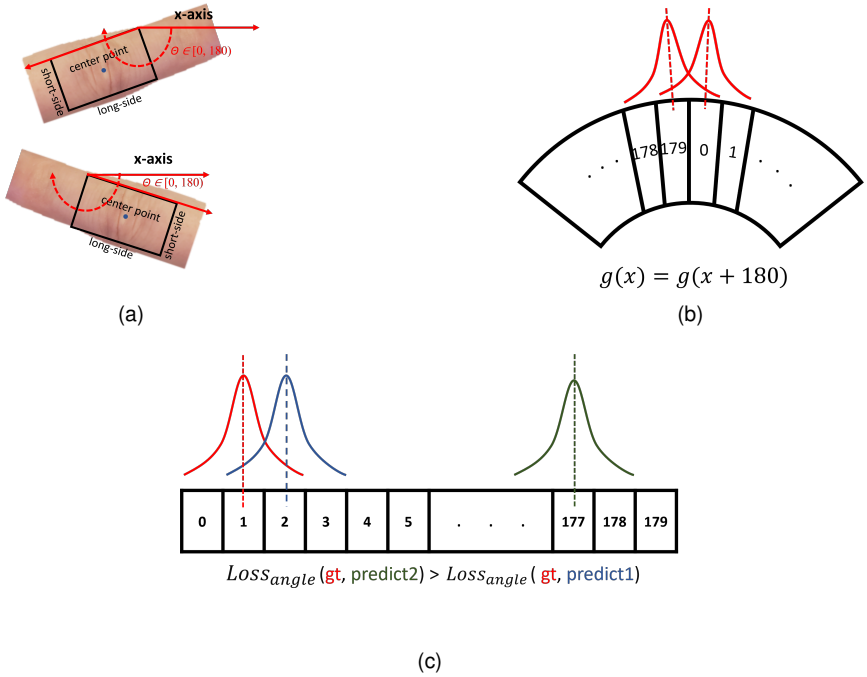


Fig. 3. Defining oriented rectangles and encoding for knuckle detection:(a) Five-parameter definition with 180° angular range. (b) CSL [36] coding with periodic Gaussian window function, (c) CSL smoothing for one-hot label during network training.

parameter method (definition in OpenCV library) [36]. Therefore, we use long side definition of bounding boxes i.e. five-parameter method with 180° angular range. This is illustrated in figure 3a and can address problems relating to the exchangeability of edges as detailed in [36].

It can be simpler to train a classification model to predict (knuckle) bounding box orientation angles as compared to the regression. A periodic coding method called referred to as Circular Smooth Label (CSL) soft coding with periodicity window function is introduced in [36] and can address two key problems for the accurate prediction of knuckle orientations: (i) it can distinguish different class relationships as shown figure 3c. When the ground truth orientation angle is 1 deg, the loss between the ground truth and 2 deg prediction is smaller than the loss between the ground truth and 177 deg prediction; (ii) it can also address the periodicity of predicted angles as also shown from illustrations in figure 3b.

Below equation illustrates the encoding of class labels using CSL for the orientation prediction, where $g(x)$ represents the window function to smooth one-hot label, r is the radius of window function while θ represents the orientation angle of the respective bounding boxes. The Gaussian window function have the periodicity of t i.e. $g(x) = g(x + t)$, and we use the t=180 deg representing as the cycle of predicting orientation labels (figure 3b).

$$CSL(x) = \begin{cases} g(x), & \theta - r < x < r + \theta \\ 0, & \text{otherwise} \end{cases} \tag{11}$$

The original YOLOv5 model uses the loss function which have three components i.e. $Loss = Loss_{bbox} + Loss_{obj} + Loss_{class}$. Since the oriented bounding box prediction is based on the YOLOv5, we also introduce angular orientation loss in this $Loss$ to enable accurate detection of knuckle ROI. Therefore, such addition of $Loss_{angle}$ to YOLOv5 loss function, the total loss function can be stated as follows:

$$Loss = Loss_{bbox} + Loss_{obj} + Loss_{class} + Loss_{angle} \tag{12}$$

$$Loss_{angle} = \sum_{i=0}^{S^2} I_{ij}^{obj} \sum_{a \in [0,180)} [\hat{P_i(a)} log(P_i(a)) + (1 - \hat{P_i(a)}) log(1 - P_i(a))] \tag{13}$$

We use the binary cross-entropy to compute the angular classification loss in (12). In above equation (13), the S^2 represents the output grid size, and the I_{ij}^{obj} represents the computation of the loss when object fall in the grid cell.

IV. Experiments and Results

This section presents thorough experimental results, using publicly available databases, to ascertain various aspects of the performance relating to its possibility for real-world applications i.e. match accuracy, complexity, and also detection accuracy. Our experiments are firstly organized to evaluate within database performance (*WithinDB*) which uses some part of the database for the training while using some other independent part of this database for the performance evaluation. It is important to note that the cross-database performance evaluation (*CrossDB*) is highly desirable to address the limitations of currently available methods as none of the finger knuckle recognition these have presented such in *CrossDB* results to the best of our knowledge. Therefore, *CrossDB* performance evaluation results are also in section IV-B. This section also includes model complexity analysis (section IV-D). Detection performance and online performance is respectively presented in section IV-F and section IV-G.

Choice of key baseline matching methods: Matching contactless finger knuckle images using a deep network-based approach has shown to offer state-of-the-art performance. The FKNet [8] has shown to offer outperforming results over several other popular networks, i.e. ResNet-50 [15], VGG-16 [40], AlexNet [41], DenseNet-121 [42], and SE-ResNet-50 [43], and therefore a judicious choice for the baseline performance comparisons. Involuntary deformations among the intra-class knuckle patterns are one of the key challenges in accurately matching *completely* contactless knuckle images. As compared to the use of convolution layer or dilated convolution [44], the deformable convolution [45] can address such local knuckle deformations by sampling at different locations and with different weights. Therefore, we also replace the convolution layer in our backbone (Fig. 1) with the deformable convolution layer, and such an attempt is referred to as DefConvNet. EfficientNet [17] is another promising but recent backbone for online knuckle matching. Both DefConvNet and EfficientNet have not been attempted in the literature, and due to their outperforming results over other strong baselines, such results are also presented for the comparisons.

It is useful to note that the important challenge or focus of this work is on developing a *completely* contactless system for real-world applications. Therefore, performance evaluation on publicly available datasets from [18] is of key importance as it is the *only database* that provides highly deformed knuckle images or includes such representative images for completely contactless applications.

A. Within Database Performance Evaluation

1) Completely Contactless Finger Knuckle Images Database (Version 3.0): This publicly available database [32] is most challenging and relevant to ascertain the effectiveness of our approach for *completely* contactless finger knuckle identification. It offers finger knuckle images from 221 different subjects and the knuckle crease deformations are significant (some even excessive to meet reasonable expectations during the deployments). It also provides two session images, and a baseline method tertiary contrast context vector (TCCV), and we use the same protocols as in this reference [18]. Two session protocols will generate 624 (104×6) genuine scores, 64272 ($104 \times 103 \times 6$) imposter scores. The network is trained using independent images from 117 ($221 - 104$) different subjects who have not provided second session images. Similarly, the first session experiments will generate 1326 (221×6) genuine match scores and 291720 ($221 \times 220 \times 6$) imposter match scores. Comparative experimental results from the first session and two session protocols, using the receiver operating characteristics (ROC) and cumulative match characteristics (CMC) are presented in Fig. 4.

Experimental results in figure 4a and 4b indicate that our approach can achieve outperforming results, not the just for the verification tasks (ROC) but also for the recognition applications (CMC). It is worth noting that this is the most challenging database but the most suitable public database for the *completely*

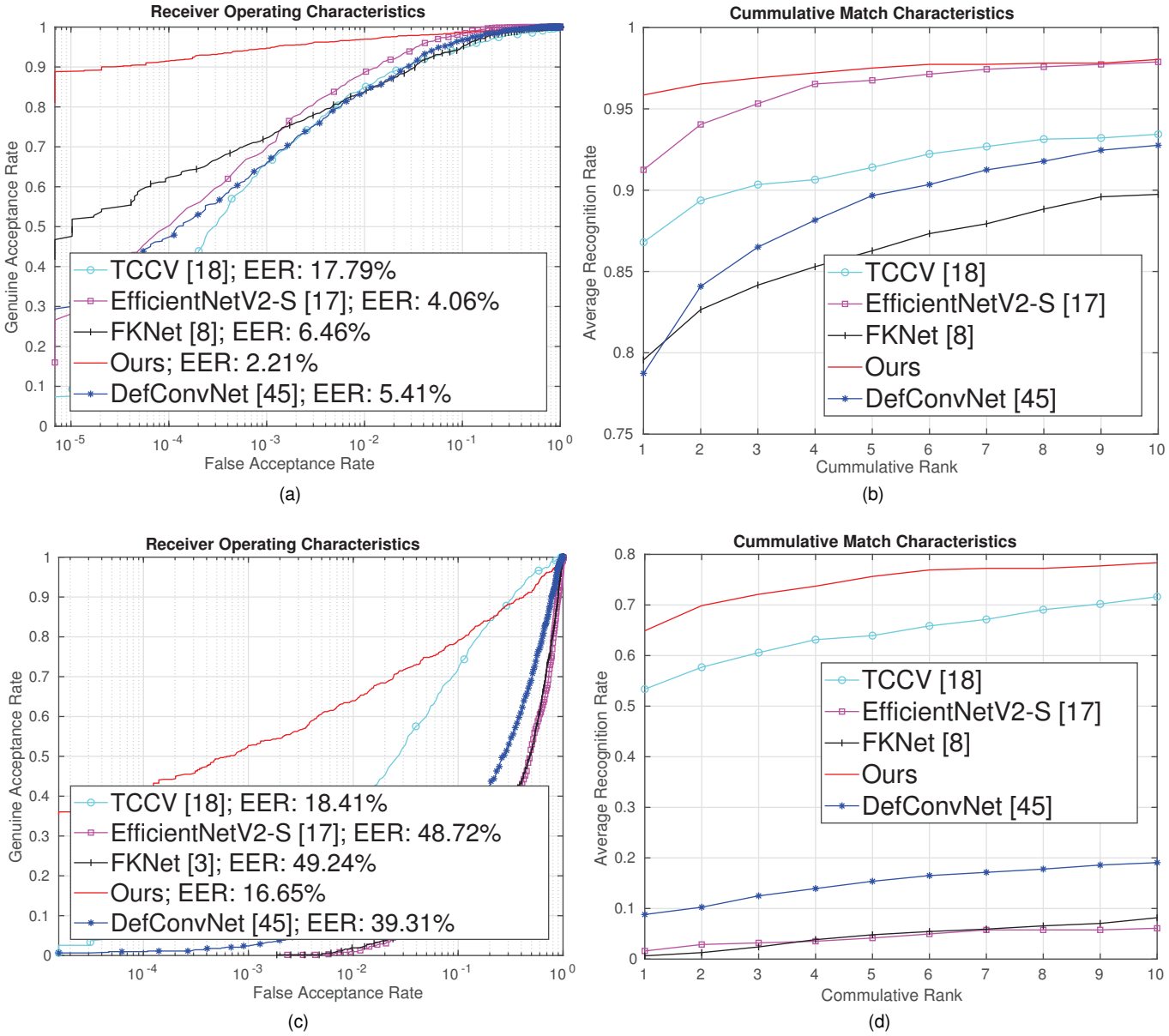


Fig. 4. Experimental results on PolyU Contactless Finger Knuckle Images Database (Version 3.0) [32] using one-session database: Comparative ROC plots (a) and CMC plots (b). Respective results from two-session database using comparative ROC plots (c) and CMC plots (d)

contactless finger knuckle identification focussed in this paper. We can also note that the EfficientNetV2-S, can offer better performance than FKNet [8] while DefConvNet can offer better performance than TCCV [18]. This may justify the choice of other baseline networks considered in our work for *completely* contactless knuckle matching. Our experiments have also indicated that if we just change the convolution layer with deformable convolution kernels, it is not adequate to accommodate finger knuckle deformations, and there is a noticeable degradation in the performance. Experimental results using the two-session database are also shown in Fig. 4. In view of the extensive degree of deformations on the two-session data, the matching accuracy is much lower than those observed from the one-session protocol. However, the proposed method can still offer the best performance than the other baseline methods considered in this work.

2) HKPolyU Contactless Hand Dorsal Images Database: This publicly available database [31] offers knuckle images from the largest number of subjects, as compared to any other public database, and was

therefore also considered for the performance evaluation. This database is acquired from 712 different subjects and each subject has provided five different images. We used index finger knuckle images and the first sample is used to fine-tune the model while the rest of the four images are used as the test samples. This kind of protocol on this database is exactly same as used in FKNet [8]. Therefore, it will generate 2848 (712×4) genuine and 2024928 ($712 \times 711 \times 4$) imposter match scores. This database has been acquired while finger poses are largely straight or fixed, and therefore the knuckle creases have much fewer deformations than those for the *completely* contactless database (Version 3.0) [32] used in the last section. It is also useful to note that it's contactless, instead of being *completely* contactless with six degrees of freedom, database as the fingers rest on a fixed surface while they are imaged using a mobile camera. It is largely due to such factors that the average performance on this database is much better than those in Fig. 4.

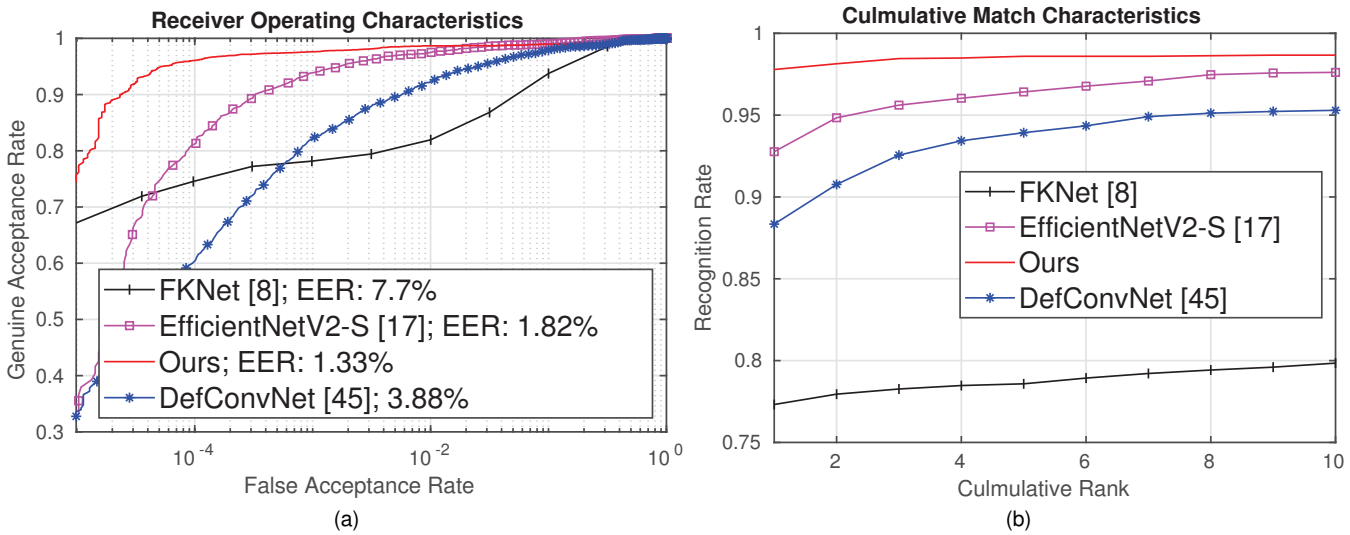


Fig. 5. Experimental results on PolyU Contactless Hand Dorsal Image Database [31] using comparative ROC plots (a) and CMC plots in (b).

It can be observed from the comparative performance shown in Fig. 5 that the proposed method can also achieve outperforming results on this database. Ours can get the best performance with a 1.33% EER, and 96.07% of average rank-one recognition rate, followed respectively by EfficientNetV2-S, DefConvNet, and FKNet. It is worth noting that the FKNet achieves worst result as compared to the other models. With classification task, the EfficientNetV2-S model is still better than the FKNet, possibly because EfficientNetV2-S have larger field of view than the FKNet. Meanwhile, EfficientNetV2-S use the Fused-MBConv [46] block while the FKNet is based on the ResNet-50 [15]. The Fused-MBConv block can enable better network learning than the residual block, and therefore EfficientNetV2-S achieves better performance than the FKNet. This is the plausible explanation for the superior performance from EfficientNetV2-S than the FKNet.

3) Contactless 2D Finger Knuckle Images from 3D Finger Knuckle Database: The HKPolyU 3D Finger Knuckle Images Database [33] is another public database, and it also provides 2D finger knuckle images (in addition to 3D finger knuckle images) which are from different subjects than those employed in the previous two databases. Therefore, this database from 190 different subjects was also employed for the performance evaluation. We use the first session 190 subjects images to fine-tune models and the second session images are used for the performance evaluation. This protocol is exactly the same as in [8] and generates 1140 (190×6) genuine and 215460 ($190 \times 189 \times 6$) imposter match scores. It can be observed from the comparative performance plots in Fig. 6 that the proposed approach can also offer better performance an EER of 1.60% and with average rank-one accuracy of 96.05%. The performance from EfficientNetV2-S is however quite close for the identification task.

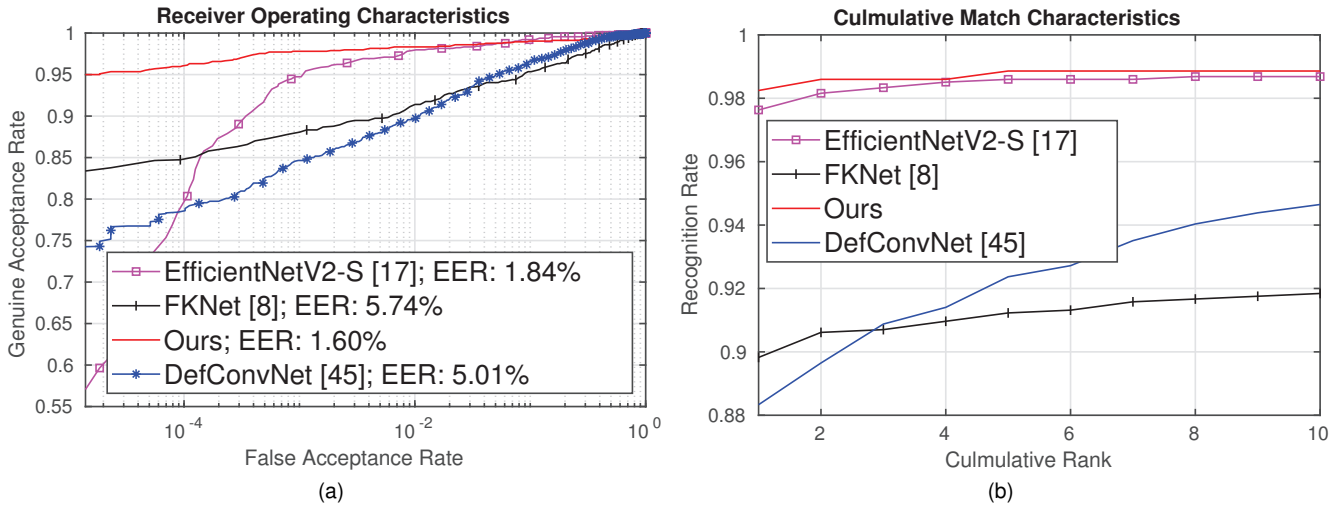


Fig. 6. Experimental results on 2D images from PolyU Contactless 3D Finger Knuckle Images Database [33] using comparative ROC plots (a) and CMC plots in (b).

B. Cross Database Performance Evaluation

This section provides results from cross-database experiments that were performed to ascertain the generalization ability of the proposed method. In this set of experiments, we used the same model that was trained for the results shown in Fig. 4 i.e. model trained using the contactless finger knuckle images database (Version 3.0) [32]. This model was used to test the performance from all finger knuckle images in the Tsinghua Finger Vein and Finger Dorsal Texture Database (THU-FVFDT3) [34] and the HKPolyU Contactless Hand Dorsal Images Database [31].

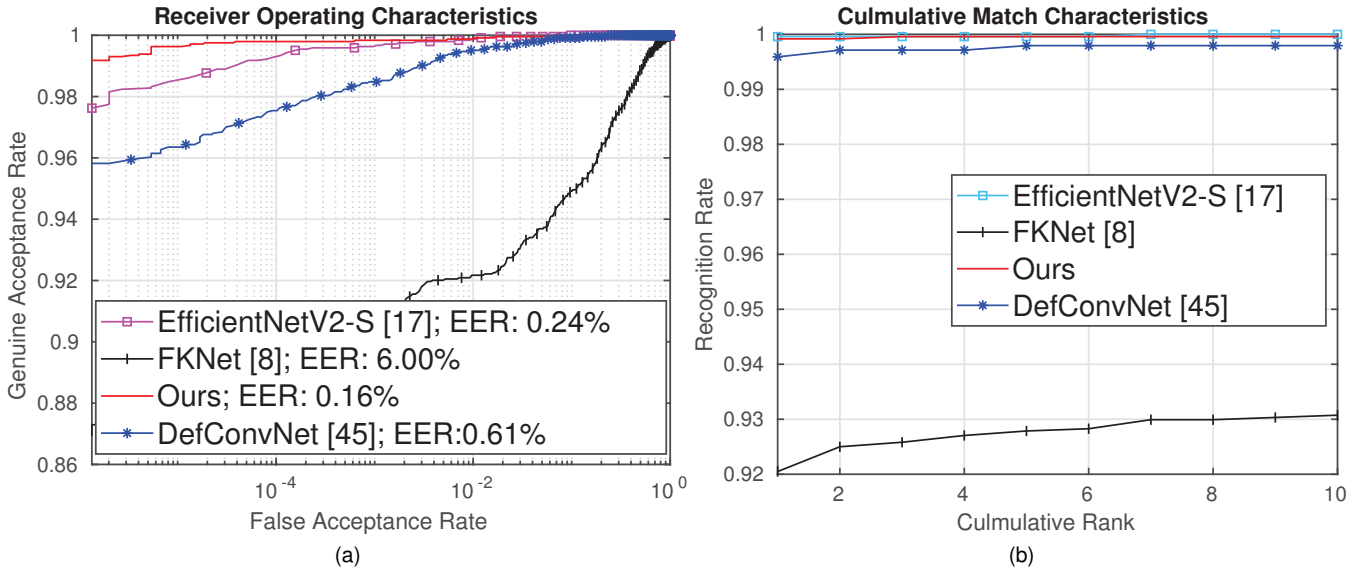


Fig. 7. Cross database experimental results on Tsinghua Finger Vein and Finger Dorsal Texture Database [34], using comparative ROC plots (a) and CMC plots in (b)

1) Tsinghua Finger Vein and Finger Dorsal Texture Database: This database [34] has been acquired from the fingers of 610 subjects, and each subject has provided 4 samples of finger dorsal images. From such finger dorsal images, we use our knuckle detection model (section III) to automatically detect and segment ROI of finger knuckle images for this test dataset. During this cross-database experiment, we generate 2440 (610×4) genuine and 1485960 $(610 \times 609 \times 4)$ imposter match scores. The results from

this set of experiments are shown in Fig. 7. Intra-class variations among the knuckle images in this dataset are small and are the reason that all models can achieve relatively high match performance. These cross-database results indicate that our approach can achieve outperforming results with EER of 0.16% and an average rank-one recognition rate of almost 100%. It can also be observed that the EfficientNetV2-S can also achieve very high performance on this database. This is largely due to the fact that this database [34] has been acquired in the indoor environment with fixed imaging conditions, such as distance, angle, and illumination, and therefore knuckle creases of fingers knuckle have higher quality and little intra-class variations. This is unlike the hand dorsal image database [31] which has been acquired in a more complex and outdoor environment with high intra-class variations.

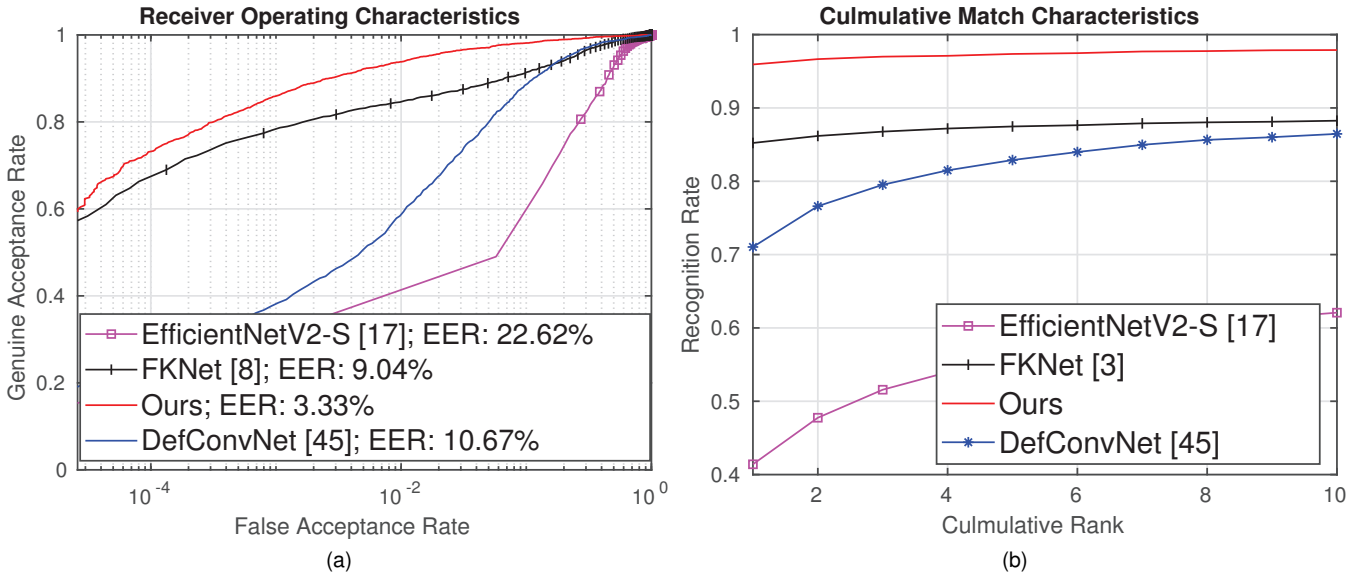


Fig. 8. Cross database experimental results on PolyU Contactless Hand Dorsal Images Database [31] using comparative ROC plots in (a) and CMC plots in (b).

2) Index Finger Knuckle of the HKPolyU Contactless Hand Dorsal Images Database: Our cross-database experiments on this database [31] use all index finger images from 712 different subjects. Therefore, such experiments will generate 3560 (712×5) genuine and 2531160 ($712 \times 711 \times 5$) imposter match scores. Such comparative experimental results are shown in Fig. 8. These results show that the proposed method can achieve outperforming results, indicating strong generalization ability from our method. However, the performance from EfficientNetV2-S has significantly degraded (EER of 22.62%) under the cross-database scenario, as compared to other models. As compared to the within-database performance that is shown in Fig. 5, performance from our method has degraded from 1.33% to 3.33% of EER. However, as for the EfficientNetV2-S model, its performance drop has been too dramatic i.e. from the 1.82% to 22.62% of EER. This can be largely due to the fact that EfficientNetV2-S is a classification network, and trained with the *soft-max* loss on a database [32] with 104 classes. However in this database there are 712 different classes or a relatively large number of unseen class numbers which can pose additional challenges during such cross-database performance evaluation.

C. 3D Finger Knuckle Images Database

In the previous part, our experimental evaluation indicated that the proposed method can achieve outperforming results on contactless 2D finger knuckle databases, both from the within- and cross-database performance evaluation. This has further encouraged us to ascertain the performance for matching contactless 3D finger knuckle images. We use a publicly available contactless 3D fingerprint database [33] and use a state-of-art method [8] and other baselines used on this database i.e. SGD [10], DenseNet [42],

and SE-ResNet [43]. The pre-processing and the match protocols used during our evaluation are exactly the same as in [8] for this database. Therefore such protocol will generate 1140 (190×6) genuine and 215, 460 $(190 \times 189 \times 6)$ imposter match scores. These experimental results are shown in Fig. 9 using comparative ROC and CMC plots. These results also indicate that our method can achieve outperforming results, both for the finger knuckle verification and identification. Our method can achieve an EER of 1.05% while the EER of the FKNet is 2.4%. We can note that the backbone used in FKNet is deeper than in our method while both of these have used the residual block. However, with the introduction of new RSIL loss, our method can offer better performance than the FKNet.

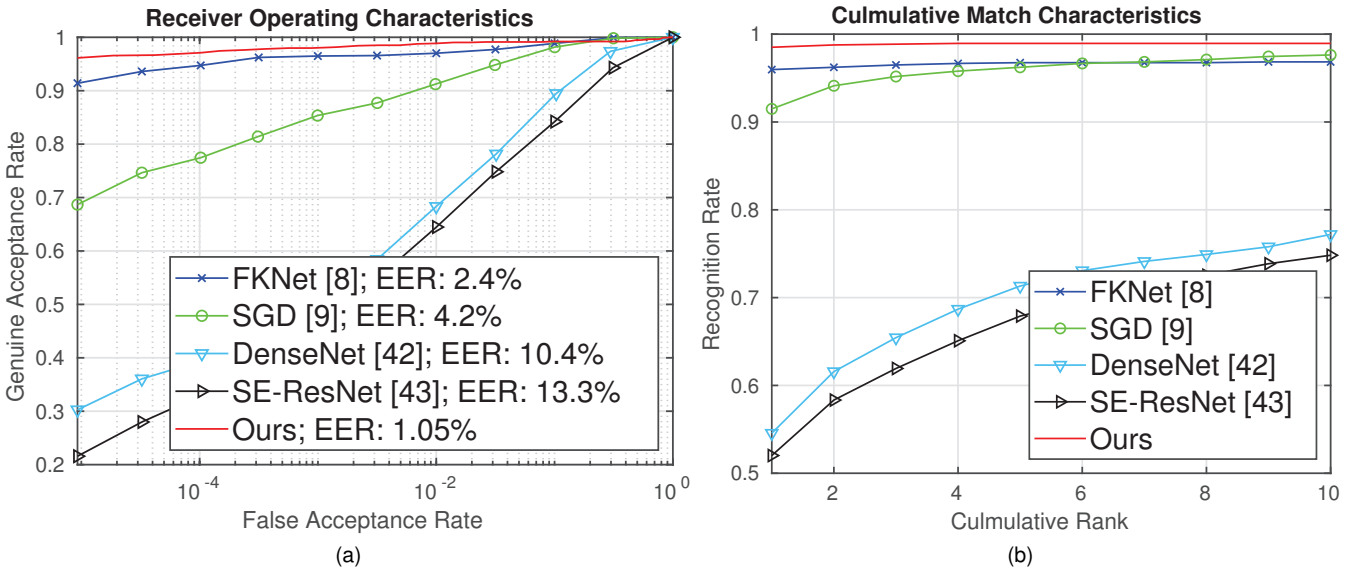


Fig. 9. Comparative experimental results for matching 3D finger knuckle images [33] using comparative ROC plots in (a) and CMC plots in (b).

D. Model Complexity Analysis

This paper is focused to develop a *completely* contactless and online finger knuckle identification system, and therefore we must choose a model that can meet the requirements of matching speed for online identification, and consider the expected trade-off between match accuracy and the speed. Therefore, this section presents a brief analysis of the complexity. Table II summarizes the computational complexity of various methods, along with the storage requirements and the corresponding feature extraction time. From the table, it can be observed that the average computational time for extracting the finger knuckle features is only 0.01957s, and storage requirements or the learnable weights are significantly smaller than the EfficientNetV2-S [17] and FKNet [8]. It is, therefore, reasonable to conclude from the experimental results in this section, that the storage complexity and time complexity of our method are significantly smaller while it can achieve the best performance from the proposed RSIL loss function.

E. Discussion

This paper has introduced a new loss function, RSIL, and has shown to offer encouraging results for matching *completely* contactless finger knuckle images. In order to ascertain the merit of this loss function over other related loss functions, i.e. triplet loss (TL) [38], and soft-shifted triplet loss (SSTL) [37], we performed additional experiments. These experiments were performed using the same backbone network (Fig. 1) on *completely* contactless finger knuckle database (version 3) [32] which has been focus of this work. These results are summarized in Fig. 10, both for the verification and identification, using

TABLE II
COMPARISON TIME AND SPACE COMPLEXITY OF DIFFERENT NEURAL NETWORK

Model	Prams (M)	Template Size	FLOPs (B)	Feature Extraction (s)
DefConvNet	0.36M	32×32	1.29B	0.02880s
EfficientNetV2-S [17]	20.18M	9×9	5.40B	0.07501s
FKNet [8]	7.28M	8×12	2.8B	0.04492s
Ours	0.46M	32×32	1.39B	0.01957s

Time complexity is the average time on the Ubuntu 22.04 with GeForce RTX 2080 GPU and I7-7800X CPU. For the STTL, the shift size is 4; for the RSIL, the shift size is 4, and add rotation with 4 angles. The FKNet and EfficientNetV2-S serial models obey their same input image size of the original paper [8] and [17], respectively.

different protocols. These comparative results also indicate the effectiveness of the proposed loss function for accurately matching *completely* contactless finger knuckle images.

We also present the respective class activation map in Fig. 11, using GRAD-CAM [47] visualization, to illustrate the salient knuckle regions contributing significantly to the output feature maps. The map is chosen such that the more red and dark regions illustrate higher importance of corresponding spatial locations. Our loss (RSIL) can account for frequent rotational and translational changes in such knuckle texture patterns and enable the model to more effectively learn such rotation and shift invariant local features. As for texture matching scenario, the key points, e.g. knuckle crease endings or the bifurcations, are more robust for matching, because these points are relatively stable to the changes in the illumination or view angle. It can be observed from the images in figure 11b that when the model is trained with the proposed loss function, then the output feature maps will emphasize (focus) on these key points and imparts higher weights on these key points as compared to the corresponding images in figure 11c or 11d. This can be more clearly visualized from a dark point in the original input knuckle images in figure 11a i.e. only our method can dominantly capture this key point information.

Our experimental results in section IV-A also suggest that EfficientV2-S model can offer better performance than FKNet [8], and even on the Tsinghua finger knuckle database (figure 7). The EfficientNetV2 model has shown to outperform ResNet on the ImageNet [48], i.e. it can extract robust features than ResNet as EfficientNetV2 replaces the residual block with inverted residual block, and uses MBConv as a block unit. As for MBConv block, it uses depth-wise convolution to decrease the learnable parameters and uses Squeeze-Excited block for the channel attention. It should be noted that the EfficientNetV2-S is deeper than the FKNet. On the contrary, the FKNet use the convolution layers of ResNet-50 up to $Conv3_x$ as the feature extract backbone. EfficientNetV2 uses more light, advanced and efficient module than ResNet. Due to these factors, the EfficientNetV2-S can perform better than the FKNet for the finger knuckle identification tasks.

The generalization capability of the RSIL is also quite good as indicated from the cross-database experimental results. As indicated from such evaluation in Fig. 8, the EER achieved from our model on hand-dorsal database is 3.33% which is although still higher as compared to within-database results in Fig. 5. However, for rest of the models, such cross-database performance degradation is huge as compared to ours. For example, on the hand dorsal database [31], the cross database EER value of DefConvNet drops from 3.88% to 10.67%, and the EfficientNetV2-S drops from 1.82% to 22.62%. In other words, such performance indicates that our RSIL loss function can alter the back propagation during training process to learn more effective model weights, which can also be noticed from the class activation maps illustrated in Fig. 11. In summary, it is reasonable to conclude that the proposed method can offer the best performance and superior generalization capability as compared to the rest of the baseline models

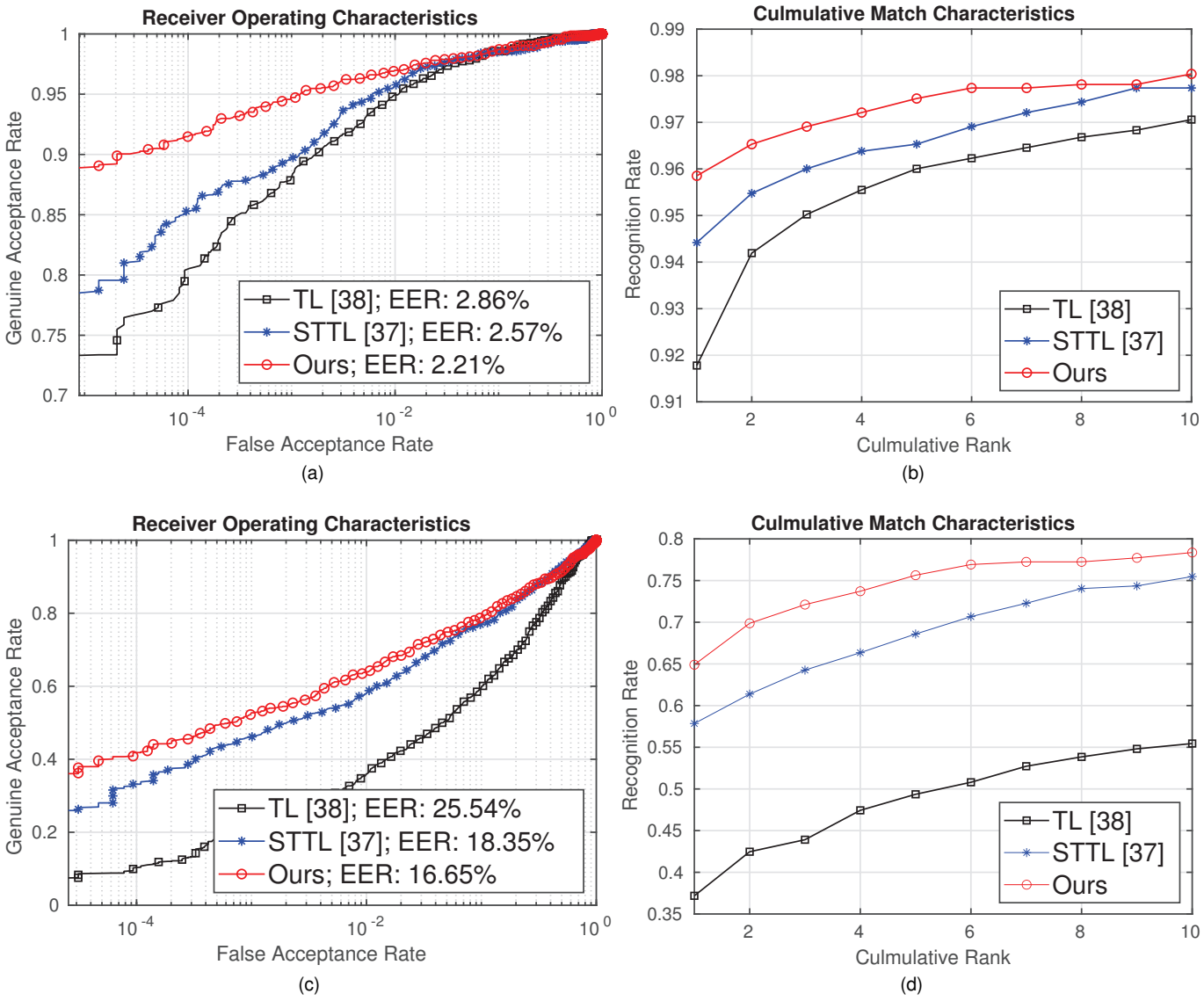


Fig. 10. Comparative match performance from our network trained network with different loss functions on PolyU Contactless Finger Knuckle Images Database (Version 3.0) [32] using one-session database: Comparative ROC plots (a) and CMC plots (b). Respective results from two-session database using comparative ROC plots (c) and CMC plots (d).

considered in this paper. Our method also offers minimal time complexity to extract the features, as can be observed from Table II, which is another reason for its suitability in *completely* contactless and real-world finger knuckle identification.

We can observe from the result in Fig. 4 that the performance of all models is relatively lower on the deformable finger knuckle dataset [32], as compared to the rest public finger knuckle datasets. As for the deformable dataset [32], the volunteers bend their fingers from $0°$ to $90°$. During such finger knuckle deformations, the texture formation from the knuckle creases is altered and this can degrade the knuckle detection and the matching performance. Firstly, we want to ascertain the changes in the detection performance with change in the extent of finger knuckle bending in this dataset (detail on two detectors, i.e. Our(x) and Our(m), are provided in Section IV-F). We present such detection performance using the first session of this dataset with 1326 (221×6) test finger knuckle images. Since each subject is provided with 6 samples, and the 6 samples represent the degree of bending of the finger in turn, roughly in the steps of $0°$, $18°$, $36°$, $54°$, $72°$, and $90°$. Therefore, we also use this data to test detector precision and recall in these samples, and these results are summarized in Table III. We set the detector confidence

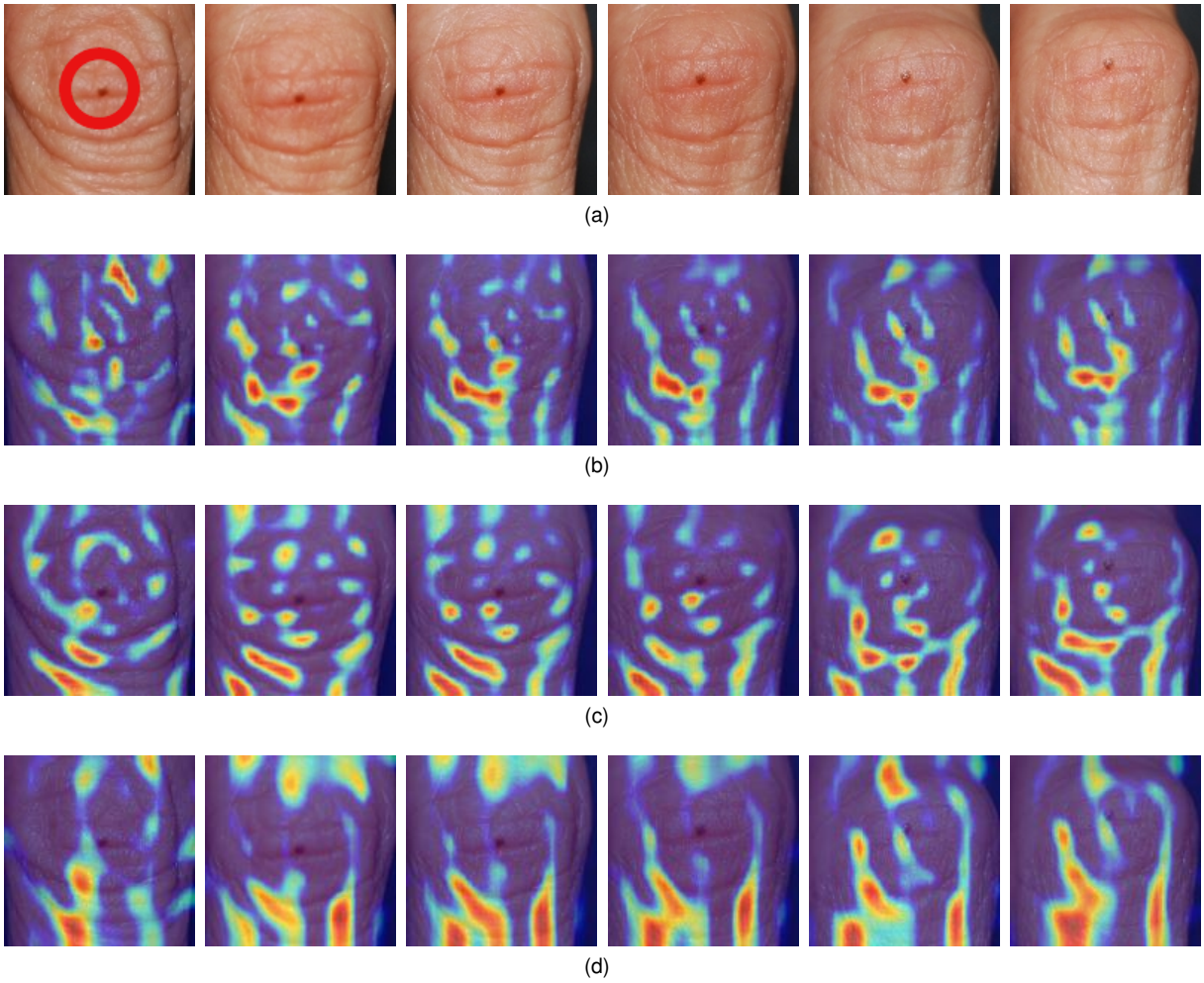


Fig. 11. Comparative class activation maps (CAM) on the images from 104th subjects samples, in PolyU Finger Knuckle Images Database (Version 3) [32], using our trained network with different loss functions (a) original input finger knuckle images, (b) results using our loss function (RSIL), (c) results using SSTL [37] and (d) triplet loss function [38].

threshold as 0.2 to test the degradation in the detection performance but during the segmentation task, we use a very low threshold to ensure the detection of each of finger knuckle ROI. It can be observed from the results in the following table that when the finger knuckle is bent by more than $54°$, there is a significant degradation in recall and precision.

TABLE III
VARIATIONS IN THE MAJOR FINGER KNUCKLE DETECTION PERFORMANCE WITH THE DEGREE OF FINGER BENDING

Sample	Model			
	Our(x)		Our(m)	
	Precision	Recall	Precision	Recall
First(0°)	0.936	0.923	0.887	0.887
Second(18°)	0.943	0.900	0.907	0.900
Third(36°)	0.913	0.896	0.904	0.855
Fourth(54°)	0.929	0.828	0.897	0.828
Fifth(72°)	0.890	0.787	0.906	0.697
Sixth(90°)	0.881	0.756	0.879	0.602

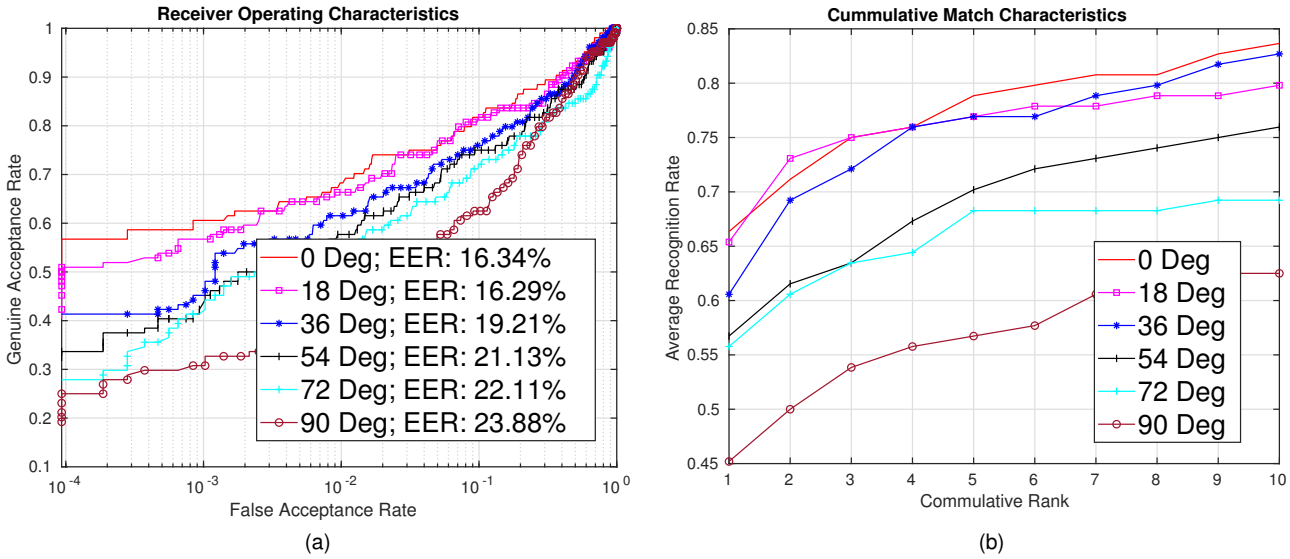


Fig. 12. Influence from finger knuckle flexion deformations for the matching performance using deformable dataset [32]: (a) ROC plots and (b) CMC plots.

Secondly, we also test the performance from our finger knuckle *recognition* model on the same dataset using such samples from each of the subjects to ascertain the changes in the match performance with the change in the finger knuckle flexion. Such result are presented in the following Fig. 12 using the ROC and CMC plots under two-session match protocols (104 genuine scores and 10712(104×103) imposter scores). These results indicate that the performance from $0°$-$36°$ is quite similar while there is remarkable degradation in the performance when the extent of finger bend is larger than $54°$.

F. Completely Contactless Finger Knuckle Detection Performance

We also developed a *completely* contactless online finger knuckle detector that can robustly detect knuckle ROI from images with complex backgrounds and in a wild. Most publicly available finger knuckle databases are acquired under a fixed background and are not sufficient to train a detector that is expected to detect knuckle ROI in the wild. Therefore, we also used a web crawler to acquire finger knuckle images from public websites [49] that provide free access to download or use [50] such images. We have downloaded 2347 images, there are 738 images without knuckles, and these images can be used as background training, and the rest of the 1609 images that contain at least one finger knuckle are the positive samples for the network model. In addition to 2347 captured images, we also include 233 images of finger joints (obtained by us) for composing our dataset, for a total of 2580 ($2347 + 233$) images. Among them, we randomly selected 2450 of them as our training set, and the remaining 100 as a testing set. The pretrained YOLOv5 model is then finetuned using these training images. We also incorporated data augmentation during the training using the popular methods i.e. flip, rotation, resize, translation and mosaicking, for the training samples. We empirically fixed r, in (11), as 6 for all the experiments in the paper.

We also performed experiments to ascertain the effectiveness of a specialized finger knuckle detector developed during this work. The merit of such specialized detector was evaluated using the (i) detection performance and (ii) average improvement in matching accuracy over key baseline methods in respective references. In order to evaluate detection performance, we employed YOLOv5x and YOLOv5m models, to predict horizontal bounding boxes, as baseline methods. Although we only used major finger knuckle during the identification in this work (simply to reduce the complexity), the detection performance was also evaluated for the detector of minor finger knuckle. Among these 2580 images, 100 images were randomly selected as the test set for this part of the evaluation and these results are summarized in Table

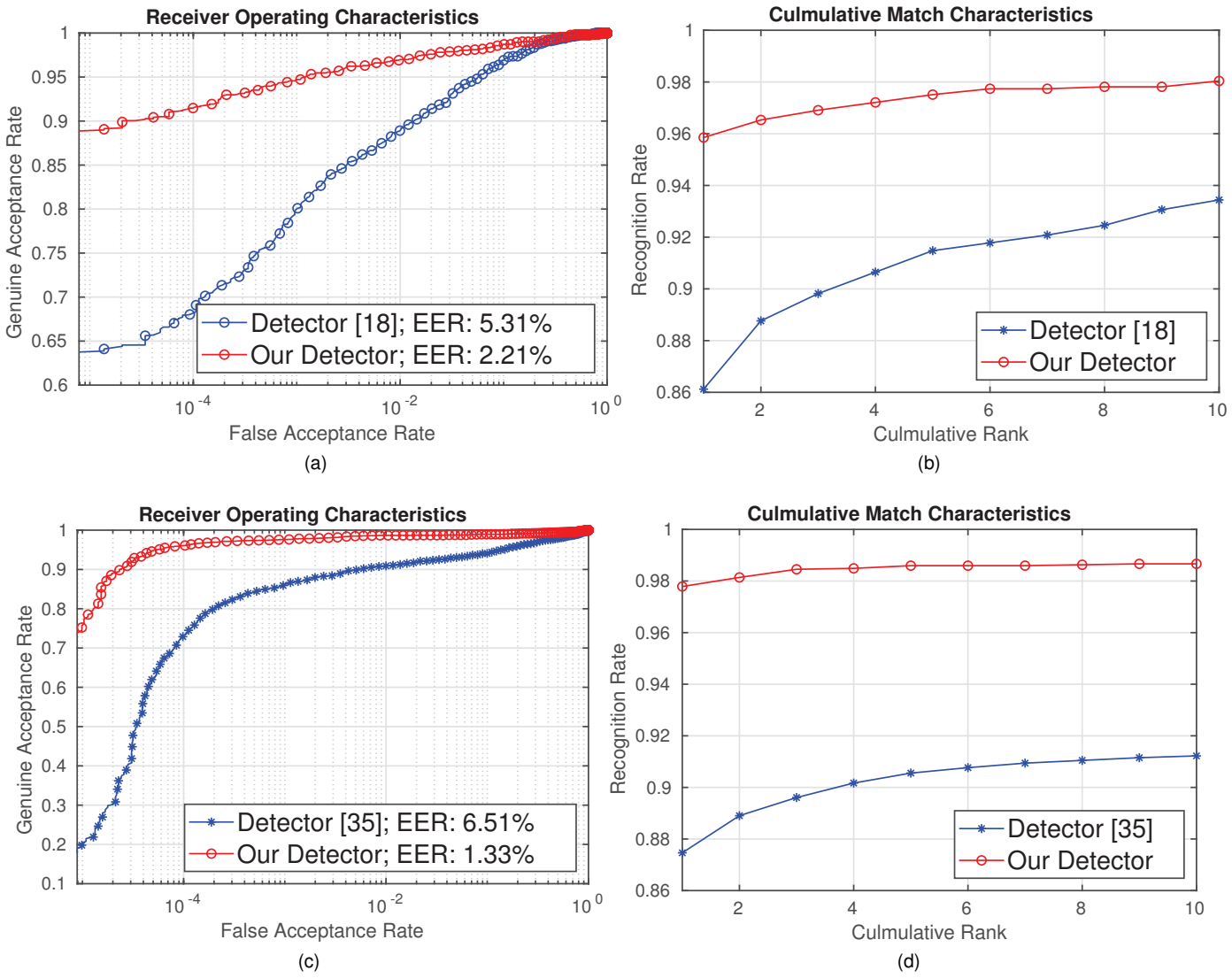


Fig. 13. Comparative match performance from our knuckle detector and using the detection method in respective references: Respective results using PolyU Contactless Finger Knuckle Images Database (Version 3.0) [32] in (a)-(b) and using PolyU Contactless Hand Dorsal Image Database [31] in (c)-(d).



Fig. 14. Sample results from our finger knuckle detector using publicly available [49] images, each of which illustrate multiple hands and complex background. The number inside the box depicts estimated orientation of respective major or major knuckle.

IV. In this table, ours(x) represent the specialized knuckle detector using YOLOv5x with CSL and the angular loss that is detailed in section III. Similarly, the ours(m) represents a lightweight version of our detector using YOLOv5m. These results indicate that the proposed knuckle detector can achieve highest mAP value of 89.9, especially, the AP of minor finger knuckle can reach to 90.1. On the contrary, without predicting an oriented bounding box, the YOLOv5x detection accuracy is lower by 1.6 mAP as compared to the performance from our(x). A video file attached with this paper provides online performance from the specialized finger knuckle detector, and none of these fingers or samples were used for training and illustrates its effectiveness for real-world deployments.

TABLE IV
COMPARISON OF THE ACCURACY OF THE YOLOV5 WITH CSL AND WITHOUT CSL FOR DETECTING FINGER KNUCKLE.

Model	Total Time/ms (1024x1024)	Number of Layers	mAP^{val} 0.5	AP of Major FK	AP of Minor FK
Our(x)	40.9ms	407	**89.9**	**89.6**	**90.1**
Our(m)	23.3ms	263	85.7	88.9	80.4
YOLOv5x	33.3 ms	407	87.3	86.5	88.0
YOLOv5m	12.1ms	263	84.8	84.5	85.1

The calculated values of mAP were measured at a detection threshold of 0.001 as well as an IOU threshold of 0.5. All the experiments are test on the GTX 2080 GPU and I7-7800X CPU, and the total time includes the inference and NMS process.

Another attempt to evaluate the effectiveness of the developed detector (and segmentation) is presented from the experimental results in Fig. 13. These plots illustrate the relative performance, on respective public databases when the finger knuckle ROI images from our detector is employed. The baseline performance represents the detected ROI images in the respective references [35], [18], using the same or our matching method. It can be observed from these results that the usage of the proposed knuckle detector can also significantly enhance the matching performance.

G. Online Contactless Finger Knuckle Identification Performance



Fig. 15. 14. Sample images from five different subjects in our video finger knuckle dataset for the online performance evaluation in this work.

We also attempted to evaluate the feasibility of the developed system for real-world applications. Therefore, we acquired a database from 52 subjects using a general-purpose smartphone. Each subject provided 15-20s video of contactless finger dorsal images with significant rotations, distance/scale changes, and under complex backgrounds (such as traffic flow, pedestrians), which can be reflective of deployment

expectations. Performance evaluation was done in a cross-database scenario and the model trained in section IV-A was employed for the performance evaluation. The minimum frame rate of these videos is 30 frames per second. We considered two scenarios for such performance evaluation: (a) under 1 image per second (Fig. 15) and (b) 6 images per second. Since the shortest video is 15s, for 1 image per second, we can get a total of 780 (52×15) finger knuckle images. Therefore by ensuring the same number of samples from each subject, 1789 (52×15) genuine and 39780 ($52 \times 51 \times 15$) imposter match scores were computed. Similarly, for the 6 images per second case, we can get 4680 ($52 \times 15 \times 6$) finger knuckle samples, which resulted in 4680 (52×90) genuine and 238680 ($52 \times 51 \times 90$) imposter match scores. The performance plots under both the considered scenarios are showed in Fig. 16. These performance plots achieve very high accuracy and indicate the suitability of the developed method and the system for a range of real-world applications.

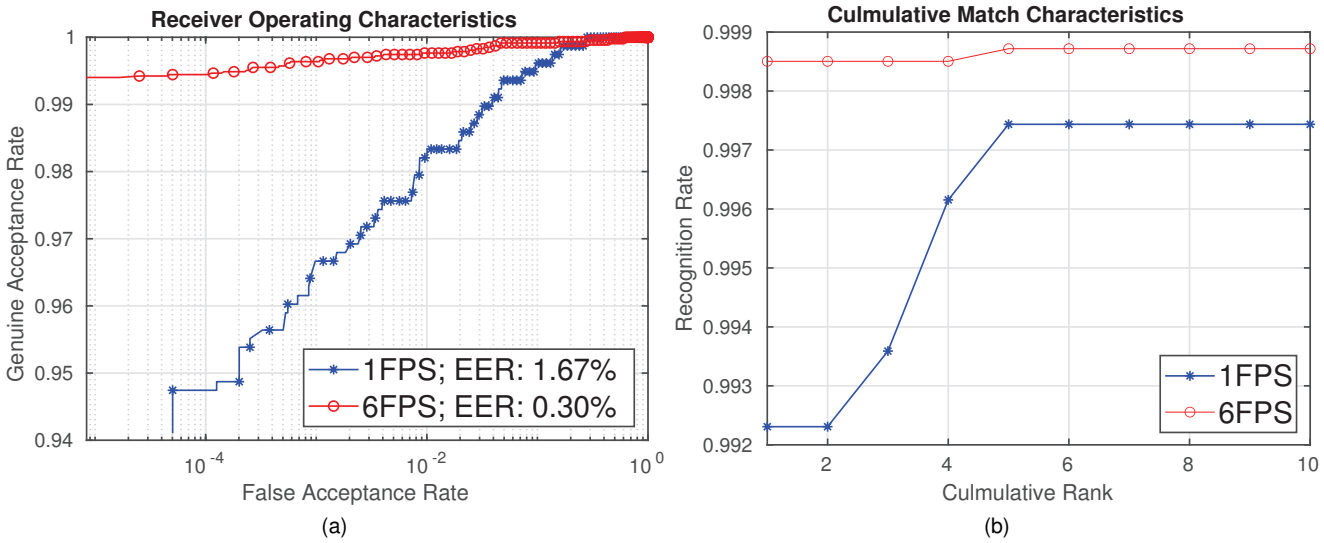


Fig. 16. Comparative performance from our completely contactless and online finger knuckle identification system using (a) ROC and CMC (b).

V. Conclusion and Further Work

This paper has proposed and developed a new framework for *completely* contactless finger knuckle identification, to meet the expectations for the real-world deployments. In order to accommodate significant intra-class variations for the completely contactless system, we proposed a new loss function (RSIL). Our experimental results presented in section IV using the proposed framework, both for the within- and cross-database experiments, indicate consistently shown outperforming results and validate the effectiveness of the framework. We also developed a robust finger knuckle detector by incorporating CSL and angular loss with a Yolov5 backbone and experimental results in section IV-F also demonstrated its effectiveness for the real-world applications. Completely contactless and online knuckle identification system was also evaluated in section IV-G and is first such system in the best of our knowledge.

Despite the impressive results and ready availability of the developed system for the real-world deployments, there are many challenges that should be addressed in the further extension of this work. The first challenge is relating to very limited availability of database. For real-world applications, the current amount of data from the finger knuckle images is quite small to allow the model to be adequately trained i.e. unlike those for the face or fingerprint recognition which have much larger dataset available for the evaluation or training. Another challenge is relating to completely contactless system is associated with the involuntarily deformations in the knuckle patterns during the access. Despite our assumption that the client has interest in access or the transaction, and therefore user will present his/her registered

fingers in the imaging volume with expected pose, some deformations in knuckle patterns are inevitable. Therefore, more effective algorithms should be designed to address such (unwanted) knuckle deformations for real-world applications. Another limitation with our system is the use of two-stage approach, where a network is trained to detect the position of finger knuckle in order to segment while other network is used to extract the finger knuckle features for matching. Development of an-end-to-end approach, e.g. multitasking, that can also incorporate features extracted for the knuckle detection, should be considered to further enhance such system. In the further extension of this work, some computationally intensive tasks can also be performed on the server side, primarily to reduce the amount of computation on mobile devices, ensuring a good online performance from mobile devices. We can also use pruning techniques and knowledge distillation for the deep neural networks to further reduce the time and space complexity, to enable its usage on a range of mobile devices.

REFERENCES

[1] M. Tistarelli, S.Z. Li, and R. Chellappa. *Handbook of Remote Biometrics: for Surveillance and Security*. Advances in Computer Vision and Pattern Recognition. Springer London, 2012.

[2] Vivek Kanhangad, Ajay Kumar, and David Zhang. A unified framework for contactless hand verification. *IEEE Transactions on Information Forensics and Security*, 6(3):1014–1027, 2011.

[3] Lin Zhang, Lei Zhang, David Zhang, and Hailong Zhu. Online finger-knuckle-print verification for personal authentication. *Pattern Recognition*, 43(7):2560–2571, 2010.

[4] Gaurav Jaswal, Amit Kaul, and Ravinder Nath. Knuckle print biometrics and fusion schemes–overview, challenges, and solutions. *ACM Computing Surveys (CSUR)*, 49(2):1–46, 2016.

[5] MorphoWave Contactless Fingerprint. https://www.idemia.com/contactless-fingerprint, Accessed July 2022.

[6] Shoichiro Aoyama, Koichi Ito, and Takafumi Aoki. A multi-finger knuckle recognition system for door handle. In *2013 IEEE Sixth International Conference on Biometrics: Theory, Applications and Systems (BTAS)*, pages 1–7. IEEE, 2013.

[7] Kenneth Okereafor, Iniobong Ekong, Ini Okon Markson, Kingsley Enwere, et al. Fingerprint biometric system hygiene and the risk of covid-19 transmission. *JMIR Biomedical Engineering*, 5(1):e19623, 2020.

[8] Kevin HM Cheng and Ajay Kumar. Deep feature collaboration for challenging 3d finger knuckle identification. *IEEE Transactions on Information Forensics and Security*, 16:1158–1173, 2020.

[9] Ajay Kumar. Importance of being unique from finger dorsal patterns: Exploring minor finger knuckle patterns in verifying human identities. *IEEE Transactions on Information Forensics and Security*, 9(8):1288–1298, 2014.

[10] Kevin HM Cheng and Ajay Kumar. Contactless biometric identification using 3d finger knuckle patterns. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 42(8):1868–1883, 2019.

[11] Wenming Yang, Zhiquan Chen, Chuan Qin, and Qingmin Liao. α-trimmed weber representation and cross section asymmetrical coding for human identification using finger images. *IEEE Transactions on Information Forensics and Security*, 14(1):90–101, 2018.

[12] Qian Zheng, Ajay Kumar, and Gang Pan. A 3d feature descriptor recovered from a single 2d palmprint image. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 38(6):1272–1279, 2016.

[13] David Zhang, Xiaoyuan Jing, and Jian Yang. *Biometric image discrimination technologies*. IGI Global, 2006.

[14] KamYuen Cheng and Ajay Kumar. Contactless finger knuckle identification using smartphones. In *2012 BIOSIG-Proceedings of the International Conference of Biometrics Special Interest Group (BIOSIG)*, pages 1–6. IEEE, 2012.

[15] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 770–778, 2016.

[16] Rajiv Kapoor, Dinesh Kumar, Anmol Garg, Anuj Sharma, et al. Completely contactless finger-knuckle recognition using gabor initialized siamese network. In *2020 International Conference on Electronics and Sustainable Communication Systems (ICESC)*, pages 867–872. IEEE, 2020.

[17] Mingxing Tan and Quoc Le. Efficientnetv2: Smaller models and faster training. In *International Conference on Machine Learning*, pages 10096–10106. PMLR, 2021.

[18] Ajay Kumar. Toward pose invariant and completely contactless finger knuckle recognition. *IEEE Transactions on Biometrics, Behavior, and Identity Science*, 1(3):201–209, 2019.

[19] Ross Girshick, Jeff Donahue, Trevor Darrell, and Jitendra Malik. Rich feature hierarchies for accurate object detection and semantic segmentation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 580–587, 2014.

[20] Ross Girshick. Fast r-cnn. In *Proceedings of the IEEE International Conference on Computer Vision*, pages 1440–1448, 2015.

[21] Shaoqing Ren, Kaiming He, Ross Girshick, and Jian Sun. Faster r-cnn: Towards real-time object detection with region proposal networks. *Advances in Neural Information Processing Systems*, 28:91–99, 2015.

[22] Joseph Redmon, Santosh Divvala, Ross Girshick, and Ali Farhadi. You only look once: Unified, real-time object detection. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 779–788, 2016.

[23] Joseph Redmon and Ali Farhadi. Yolo9000: better, faster, stronger. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 7263–7271, 2017.

[24] Joseph Redmon and Ali Farhadi. Yolov3: An incremental improvement. *arXiv preprint arXiv:1804.02767*, 2018.

[25] Alexey Bochkovskiy, Chien-Yao Wang, and Hong-Yuan Mark Liao. Yolov4: Optimal speed and accuracy of object detection. *arXiv preprint arXiv:2004.10934*, 2020.

[26] Wei Liu, Dragomir Anguelov, Dumitru Erhan, Christian Szegedy, Scott Reed, Cheng-Yang Fu, and Alexander C Berg. Ssd: Single shot multibox detector. In *European Conference on Computer Vision*, pages 21–37. Springer, 2016.

[27] Ying Xin, Guanzhong Wang, Mingyuan Mao, Yuan Feng, Qingqing Dang, Yanjun Ma, Errui Ding, and Shumin Han. Pafnet: An efficient anchor-free object detector guidance. *arXiv preprint arXiv:2104.13534*, 2021.

[28] Ultralytics. Yolov5. https://github.com/ultralytics/yolov5, 18 May 2020.

[29] Daksh Thapar, Gaurav Jaswal, and Aditya Nigam. Fkimnet: a finger dorsal image matching network comparing component (major, minor and nail) matching with holistic (finger dorsal) matching. In *2019 International Joint Conference on Neural Networks (IJCNN)*, pages 1–8. IEEE, 2019.

[30] The HKPolyU Contactless Finger Knuckle Images Database (V-1.0):. http://www4.comp.polyu.edu.hk/~csajaykr/fn1.htm.

[31] The HKPolyU Contactless Hand Dorsal Images Database:. http://www4.comp.polyu.edu.hk/~csajaykr/knuckleV2.htm.

[32] The HKPolyU Contactless Finger Knuckle Images Database (Version 3.0) :. https://www4.comp.polyu.edu.hk/~csajaykr/fn2.htm.

[33] The HKPolyU 3D Finger Knuckle Images Database:. https://www4.comp.polyu.edu.hk/~csajaykr/3DKnuckle.htm.

[34] Tsinghua University Finger Vein and Finger Dorsal Texture Database (THU-FVFDT3):. https://www.sigs.tsinghua.edu.cn/labs/vipl/thu-fvfdt.html.

[35] Ajay Kumar and Zhihuan Xu. Personal identification using minor knuckle patterns from palm dorsal surface. *IEEE Transactions on Information Forensics and Security*, 11(10):2338–2348, 2016.

[36] Xue Yang and Junchi Yan. Arbitrary-oriented object detection with circular smooth label. In *European Conference on Computer Vision*, pages 677–694. Springer, 2020.

[37] Yang Liu and Ajay Kumar. Contactless palmprint identification using deeply learned residual features. *IEEE Transactions on Biometrics, Behavior, and Identity Science*, 2(2):172–181, 2020.

[38] Florian Schroff, Dmitry Kalenichenko, and James Philbin. Facenet: A unified embedding for face recognition and clustering. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 815–823, 2015.

[39] Kaiming He, Georgia Gkioxari, Piotr Dollár, and Ross Girshick. Mask r-cnn. In *Proceedings of the IEEE International Conference on Computer Vision*, pages 2961–2969, 2017.

[40] Karen Simonyan and Andrew Zisserman. Very deep convolutional networks for large-scale image recognition. *arXiv preprint arXiv:1409.1556*, 2014.

[41] Alex Krizhevsky, Ilya Sutskever, and Geoffrey E Hinton. Imagenet classification with deep convolutional neural networks. *Advances in Neural Information Processing Systems*, 25, 2012.

[42] Gao Huang, Zhuang Liu, Laurens Van Der Maaten, and Kilian Q Weinberger. Densely connected convolutional networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 4700–4708, 2017.

[43] Jie Hu, Li Shen, and Gang Sun. Squeeze-and-excitation networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 7132–7141, 2018.

[44] Fisher Yu and Vladlen Koltun. Multiscale context aggregation by dilated convolutions. *arXiv preprint arXiv:1511.07122*, 2015.

[45] Xizhou Zhu, Han Hu, Stephen Lin, and Jifeng Dai. Deformable convnets v2: More deformable, better results. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 9308–9316, 2019.

[46] Suyog Gupta and Mingxing Tan. Efficientnet-edgetpu: Creating accelerator-optimized neural networks with automl. https://ai.googleblog.com/2019/08/efficientnet-edgetpu-creating.html, 2019.

[47] Ramprasaath R Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, and Dhruv Batra. Grad-cam: Visual explanations from deep networks via gradient-based localization. In *Proceedings of the IEEE International Conference on Computer Vision*, pages 618–626, 2017.

[48] Olga Russakovsky, Jia Deng, Hao Su, Jonathan Krause, Sanjeev Satheesh, Sean Ma, Zhiheng Huang, Andrej Karpathy, Aditya Khosla, Michael Bernstein, et al. Imagenet large scale visual recognition challenge. *International Journal of Computer Vision*, 115(3):211–252, 2015.

[49] Unsplash. Unsplash. https://unsplash.com/.

[50] Unsplash. Unsplash.com. "Unsplash License", Accessed 11 January 2022.